# AGAP: A GPS-Based Mobile Emergency Alert System for Immediate Hospital / Ambulance Dispatch

## MODULE 05 — Emergency Readiness and Permission Status

### Module Description

**Description** Checks whether a citizen's phone and account are ready to send an emergency alert. It shows a checklist (location, notifications, internet connection, camera, motion sensor, emergency contacts, verified phone number and medical profile), rates each item, and gives an overall status of Ready, Limited or Not Ready. Each item has a button to fix it.

**Purpose** Find problems, such as location permission being off, before an emergency happens, so the citizen can fix them while there's time. The checklist never blocks an emergency report.

### Input

**AS-IS Input:**

- Resident's own judgment of whether their phone worksSource bulletins (PAGASA, river gauge reading, DepEd / LGU orders)
- Phone signal and load at the time of the emergency

**TO-BE Input:**

- Location permission and whether location services are on
- Notification permission
- Internet connection status
- Camera permission
- Motion sensor availability
- Device ID
- Account status: emergency contacts, phone verification, medical consent and profile

### Technology Used

**AS-IS Technologies:**

- Resident's personal mobile phone
- Telephone / MDRRMO hotline (used only when the emergency happens)

**TO-BE Technologies:**

- AGAP Citizen mobile app (Expo, React Native, Expo Router)
- Expo Location, Expo Notifications, Expo Camera, Expo Sensors (permission and sensor status)
- Network status check (online, limited, offline)
- NestJS REST API (GET and PUT /api/v1/users/me/readiness) with JWT sign-in check
- Readiness rules on the server, with the same rules in the app for use offline
- PostgreSQL (device_readiness_reports table)

## Process Documentation

### AS-IS Process (Existing Process)

| Steps | Process Description |
|---|---|
| 1 | No readiness check takes place before an emergency. |
| 2 | When an emergency happens, the resident tries to call the hotline from their phone. |
| 3 | Problems such as no load, no signal or a dead battery are discovered only at that moment. |
| 4 | The resident looks for another phone or asks a neighbor to call. |
| 5 | Help is delayed until a working line is found. |

### AS-IS Outputs

| No. | Output Produced by the Existing Process |
|---|---|
| 1 | Phone problems discovered only during the emergency |
| 2 | Delayed call for help while a working phone is found |

### TO-BE Process (Proposed Automated Process)

| Step | Process Description | Step Type |
|---|---|---|
| 1 | The signed-in citizen opens Profile, then Emergency Readiness. The check also runs again each time the screen comes back into view. | User Action |
| 2 | The app reads the location permission and whether location services are on, plus the notification and camera permissions, motion sensor availability and internet connection. | Processing |
| 3 | The app sends this device report with the device ID to the backend. | Integration |
| 4 | The backend combines the device report with the account status (contacts, verified phone, medical consent and profile) and rates each item Ready, Limited or Action Needed. | Processing |
| | | |
| 5 | The backend works out the overall status. Not Ready if any Essential item (location, notifications, connection) needs action; Limited if any Recommended item (contacts, verified phone) isn't ready; otherwise Ready. Optional items (camera, motion sensor, medical profile) never lower it. | Validation |
| 6 | The backend saves the latest report for that device, replacing the previous one. | Database |
| 7 | The app shows the overall status and the checklist, with Essential items first. | Processing |

| | | |
|---|---|---|
| **8** | Without internet, the app applies the same rules on the phone and shows the result so the citizen still sees what to fix. | Processing |
| **9** | The citizen taps an item's action: allow a permission, open phone settings, add a contact, open Profile, manage consent, or set up the medical profile. | User Action |
| **10** | After the fix, the citizen pulls to refresh, and the status is checked and saved again. A button to call the emergency hotline is always shown on the screen. | Processing |

| **TO-BE Process Outputs** | |
|---|---|
| **No.** | **Output Produced by the Proposed Process** |
| **1** | Overall readiness status (Ready, Limited or Not Ready) |
| **2** | Checklist with the status and reason for each item |
| **3** | Saved readiness report for the device in PostgreSQL |
| **4** | Fix actions that take the citizen to the right permission or screen |

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